

A monotone comparison-tensor obstruction for a small exceptional die

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Status. The structural theorem below has a complete elementary proof. Together with the independently checked arithmetic result in the companion note, it gives a universal $n - 1$ per-die lower bound. This bound is attained for $n = 4$ by the separately certified construction. No manuscript source was edited.

Let D_0 have c faces in an $(m + 1)$ -die permutation-fair collection. For its faces $q = 1, \dots, c$, in increasing order, put $p_j(q) = \Pr(D_j < D_0(q))$, $1 \leq j \leq m$. These are nondecreasing rational-valued functions of q . Write $\mathbb{E}$ for uniform averaging over q . Permutation fairness on every subset gives

$$\mathbb{E} \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]). \quad (1)$$

Only these moments and the common monotone order are needed below.

Theorem 1 (Reduction of a short monotone mixture). *Suppose $p_1, \dots, p_m : \{1, \dots, c\} \rightarrow [0, 1]$ are nondecreasing and satisfy (1). If $c < m$, there is an index j_0 such that*

$$\frac{1}{c} \sum_{q=1}^c p_{j_0}(q)^a = \frac{1}{a + 1} \quad (0 \leq a \leq m).$$

In particular, the c numbers $p_{j_0}(q)$ form an equal-weight interval quadrature of degree m . They are rational if all $p_j(q)$ are rational.

Proof. For real-valued functions on the c -point probability space, use the inner product $\langle f, g \rangle = \mathbb{E}(fg)$. Let $v_j = p_j - 1/2$ and $\rho = 1/12$. The first two orders of (1) give $\mathbb{E}v_j = 0$ and $\langle v_i, v_j \rangle = \rho$ for $i \neq j$. Thus the Gram matrix has the form

$$G = D + \rho \mathbf{1}\mathbf{1}^T, \quad D = \text{diag}(d_1, \dots, d_m), \quad d_j = \|v_j\|^2 - \rho.$$

All v_j lie in a $(c - 1)$ -dimensional space, so $\text{rank } G \leq c - 1$. Consequently $\text{rank } D \leq c$. Since $m > c$, some $d_{j_0} = 0$. Set $p = p_{j_0}$ and $v = p - 1/2$. For every $j \neq j_0$, the function $e_j = p_j - p$ satisfies

$$\langle e_j, 1 \rangle = \langle e_j, v \rangle = 0, \quad \|e_j\|^2 = d_j \geq 0.$$

Moreover $\langle e_i, e_j \rangle = 0$ for distinct $i, j \neq j_0$. Thus the nonzero e_j are mutually orthogonal and are orthogonal to both 1 and the nonconstant v . There are at most $c - 2$ of them. Let k be their number and put $r = m - k$. Exactly r of the p_j equal p , and $r \geq m - c + 2$.

Let t be the number of distinct values taken by p . Products of distinct members of the r -element equal group show that

$$\mathbb{E}p^a = \frac{1}{a+1} \quad (0 \leq a \leq r).$$

For an exceptional index j and $0 \leq a < r$, choosing a distinct members of that group gives

$$\mathbb{E}(p^a p_j) = \frac{1}{a+2} = \mathbb{E}p^{a+1}, \quad \mathbb{E}(p^a e_j) = 0.$$

If $t \geq r$, the r functions $1, p, \dots, p^{r-1}$ are linearly independent. The k nonzero orthogonal functions e_j lie in their common orthogonal complement, so $k \leq c - r$. This contradicts $k + r = m > c$. Hence $t < r$.

Polynomial interpolation on the t values of p now gives

$$\mathbb{E}(e_j \mid p) = 0 \tag{2}$$

for every exceptional j . Since p is nondecreasing, each level set of p is a consecutive block of the ordered q 's. On such a block B , every $e_j = p_j - p$ is nondecreasing. Chebyshev's sum inequality and (2) therefore give

$$\mathbb{E}(e_i e_j \mid B) \geq 0.$$

Its unconditional expectation is zero by the already proved orthogonality, so it is zero on every block. For two nondecreasing real sequences on a finite interval, equality in this covariance inequality implies that at least one sequence is constant: indeed,

$$\text{Cov}_B(f, g) = \frac{1}{2|B|^2} \sum_{a, b \in B} (f(a) - f(b))(g(a) - g(b)),$$

and if both sequences are nonconstant, the summand from the first and last positions is strictly positive. By (2), a constant e_j on B is zero. Thus, on each level block of p , at most one exceptional function e_j is nonzero.

For any subset $S \subseteq [m]$, the product $\prod_{j \in S} p_j$ consequently has, on a level block $p = x$, either the constant value $x^{|S|}$ or the form $x^{|S|} + x^{|S|-1}e_j$. In either case its conditional expectation is $x^{|S|}$. Hence

$$\mathbb{E}p^{|S|} = \mathbb{E} \prod_{j \in S} p_j = \frac{1}{|S| + 1}.$$

Taking sets of every size from 0 to m proves the theorem. $\square$

Remark 2 (Positive atom weights and repeated joint rows). *The structural proof also holds for arbitrary strictly positive weights on the c ordered atoms, with the conclusion $\mathbb{E}p^a = 1/(a + 1)$. All dimension and interpolation arguments are unchanged. On a level block, the only modification is the weighted covariance identity*

$$\text{Cov}(f, g) = \frac{1}{2} \sum_{a, b} w_a w_b (f(a) - f(b))(g(a) - g(b)),$$

where the positive conditional weights sum to one. Its equality case for non-decreasing functions is the same as above. Consequently, if a uniform comparison tensor has fewer than m distinct joint rows, merging equal consecutive rows and then undoing the merge gives a scalar quadrature on its original number of equal atoms, including their repetitions.

Corollary 3 (Universal per-die bound). *Every die in every permutation-fair collection of $n \geq 2$ dice has at least $n - 1$ faces.*

Proof. Put $m = n - 1$ and suppose that a die has $c < m$ faces. By Theorem 1, there is a rational equal-weight interval quadrature with c nodes and degree m . The arithmetic theorem proved in the companion note `rational_designs/two_adic_lower_bound.tex` says that such a rule has at least $2^{\lfloor m/2 \rfloor}$ nodes. For $m \geq 6$, this is at least m , a contradiction. For $2 \leq m \leq 5$, only $c \leq 4$ need be considered. A one-node rule cannot have degree two. For $c = 2, 3, 4$, map the nodes to $[-1, 1]$. Since $m > c$, their first c power sums would equal those of the uniform measure, multiplied by c . Newton identities force their node polynomial, up to scalar, to be respectively

$$3x^2 - 1, \quad x(2x^2 - 1), \quad 45x^4 - 30x^2 + 1.$$

None splits into rational linear factors: the first two have visibly irrational roots, and the squared roots of the last are $(5 \pm 2\sqrt{5})/15$. Thus no such rational quadrature exists. For $m = 1$, the desired bound is just the requirement that each die have a face. $\square$

Independent classical alternative. Without the new arithmetic theorem, the structural result already gives $c \geq n - 1$ for all sufficiently large n : uniform-interval equal-weight quadrature of degree m requires $\Omega(m^2)$ nodes, contradicting $c < m$. The classical Bernstein theorem and the finite calculations below can also remove the remaining small cases.

Sources for the classical input. The lower quadratic order is stated in Kuijlaars, *The minimal number of nodes in Chebyshev type quadrature formulas* (1993), DOI 10.1016/0019-3577(93)90007-L; it also follows from Gilboa–Peled, *Chebyshev-type quadratures for doubling weights*, arXiv:1507.01505. A stronger classical theorem of Bernstein says that an equal-weight c -node uniform-interval quadrature of degree at least c can have all nodes real only for $c \in \{1, 2, 3, 4, 5, 6, 7, 9\}$. A primary paper explicitly recording this theorem is Gautschi–Monegato, *On Optimal Chebyshev-Type Quadratures*, Numerische Mathematik 28 (1977), 59–67, author-hosted PDF. Its input is not needed if the parallel rational quadrature lower bound is used.

Small rational cases. After mapping nodes to $[-1, 1]$, a c -node degree- c quadrature has power sums $s_a = 0$ for odd a and $s_a = c/(a + 1)$ for even $a \leq c$. Newton identities uniquely determine the monic polynomial of its nodes. For the exceptional c permitted by Bernstein, the nontrivial factors are

c	monic polynomial, up to a nonzero scalar
2	$3x^2 - 1$
3	$x(2x^2 - 1)$
4	$45x^4 - 30x^2 + 1$
5	$x(72x^4 - 60x^2 + 7)$
6	$105x^6 - 105x^4 + 21x^2 - 1$
7	$x(6480x^6 - 7560x^4 + 2142x^2 - 149)$
9	$x(22400x^8 - 33600x^6 + 15120x^4 - 2280x^2 + 53)$.

Each displayed nonconstant factor other than x is irreducible over $\mathbb{Q}$ (verified exactly by the accompanying SymPy script), so none has all rational roots. For a short stand-alone proof, finite rational-root tests suffice, and their complete integer certificates can be recorded. Together with Bernstein,

this rules out rational equal-weight degree- m quadrature with $2 \leq c \leq m$, implying $c \geq n - 1$ for all $n \geq 3$. The case $c = 1$, $m \geq 2$ fails already by the first two moments.

Interpretation and remaining obstacle. The theorem does not prove an exponential per-die bound: it applies only when $c < n - 1$. The arithmetic rational-design obstruction, however strong, therefore cannot simply be applied to every die. Extending the reduction past this low-rank threshold would require new information. The existing three-die example with a two-face die also shows that an unrestricted rational-quadrature reduction is false: its two comparison functions give a rational product mixture, but there is no rational equal-weight two-node degree-two interval rule.